Chapter 5. Introduction to Factorial Designs

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1 Introduction

In a **factorial design**, every complete replicate investigates all possible combinations of the levels of the factors.

- **Main effect**: change in response produced by a change in the level of a factor.
- **Interaction**: the difference in response between the levels of one factor is not the same at all levels of the other factor.

Example: Main effects and interaction in a 2×2 layout

Responses: $(A^-, B^-) = 20$, $(A^+, B^-) = 40$, $(A^-, B^+) = 30$, $(A^+, B^+) = 52$.

$$\begin{aligned} A &= \bar{y}_{A^+} - \bar{y}_{A^-} = \frac{40+52}{2} - \frac{20+30}{2} = 46 - 25 = 21, \\ B &= \frac{30+52}{2} - \frac{20+40}{2} = 41 - 30 = 11, \\ AB &= \frac{1}{2}[(\text{A effect at } B^+) - (\text{A effect at } B^-)] \\ &= \frac{1}{2}[(52-30) - (40-20)] = \frac{1}{2}(22-20) = 1. \end{aligned}$$

If instead $(A^+, B^+) = 12$: A effect at B^- is 20, at B^+ is $12-30 = -18$; $AB = \frac{1}{2}(-18-20) = -19$. A large interaction makes the main effect $A = \frac{40+12}{2} - 25 = 1$ misleading.

Advantages of factorial experiments

1. More efficient than one-factor-at-a-time experiments.
2. All data are used in computing *both* effects (all observations are used in determining the average effect of A and the average effect of B).
3. Some information is provided on possible interaction between the factors; one-factor-at-a-time cannot detect interaction and may lead to wrong conclusions.

2 The Two-Factor Factorial Design

2.1 Battery design example

Factor A = plate material type ($a = 3$), factor B = temperature ($b = 3$: 15, 70, 125°F), $n = 4$ replicates, all $abn = 36$ runs in random order (a CRD on the ab cells). Response: battery life (hours).

Material	15°F	70°F	125°F	$y_{i\cdot\cdot}$
1	130, 155 74, 180	34, 40 80, 75	20, 70 82, 58	998
	$y_{11\cdot} = 539$	229	230	
2	150, 188 159, 126	136, 122 106, 115	25, 70 58, 45	1300
	623	479	198	
3	138, 110 168, 160	174, 120 150, 139	96, 104 82, 60	1501
	576	583	342	
$y_{\cdot j\cdot}$	1738	1291	770	$y_{\cdots} = 3799$

Questions: (1) What effects do material type and temperature have on life? (2) Is there a material giving long life regardless of temperature (a *robust* product)?

2.2 Model and hypotheses

Definition: Two-factor factorial (fixed effects)

Effects model:

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}, \quad i = 1,\dots,a;\ j = 1,\dots,b;\ k = 1,\dots,n,$$

$\sum_i \tau_i = 0$, $\sum_j \beta_j = 0$, $\sum_i (\tau\beta)_{ij} = \sum_j (\tau\beta)_{ij} = 0$, $\varepsilon_{ijk} \overset{\text{iid}}{\sim} N(0,\sigma^2)$.
Means model: $y_{ijk} = \mu_{ij} + \varepsilon_{ijk}$, $\mu_{ij} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij}$.

Three hypotheses:

1. $H_0 : \tau_1 = \cdots = \tau_a = 0$ vs H_1: at least one $\tau_i \neq 0$;
2. $H_0 : \beta_1 = \cdots = \beta_b = 0$ vs H_1: at least one $\beta_j \neq 0$;
3. $H_0 : (\tau\beta)_{ij} = 0$ for all i, j vs H_1: at least one $(\tau\beta)_{ij} \neq 0$.

2.3 Least squares estimation

Derivation

$S(\theta) = \sum_i \sum_j \sum_k \left(y_{ijk} - \mu - \tau_i - \beta_j - (\tau\beta)_{ij}\right)^2$.

$$\begin{aligned}
\left.\frac{\partial S}{\partial \mu}\right|_{\hat\theta} &= -2\sum_i\sum_j\sum_k \left(y_{ijk} - \hat\mu - \hat\tau_i - \hat\beta_j - \widehat{(\tau\beta)}_{ij}\right) = 0 \\
&\Rightarrow\ abn\hat\mu = y_{\cdots} \ \Rightarrow\ \hat\mu = \bar y_{\cdots}, \\
\left.\frac{\partial S}{\partial \tau_i}\right|_{\hat\theta} &= -2\sum_j\sum_k \left(y_{ijk} - \hat\mu - \hat\tau_i - \hat\beta_j - \widehat{(\tau\beta)}_{ij}\right) = 0 \\
&\Rightarrow\ bn\hat\tau_i = y_{i\cdot\cdot} - bn\hat\mu \ \Rightarrow\ \hat\tau_i = \bar y_{i\cdot\cdot} - \bar y_{\cdots}, \\
\left.\frac{\partial S}{\partial \beta_j}\right|_{\hat\theta} &= 0 \ \Rightarrow\ \hat\beta_j = \bar y_{\cdot j\cdot} - \bar y_{\cdots}, \\
\left.\frac{\partial S}{\partial (\tau\beta)_{ij}}\right|_{\hat\theta} &= -2\sum_k \left(y_{ijk} - \hat\mu - \hat\tau_i - \hat\beta_j - \widehat{(\tau\beta)}_{ij}\right) = 0 \\
\Rightarrow\ \widehat{(\tau\beta)}_{ij} &= \bar y_{ij\cdot} - \bar y_{\cdots} - (\bar y_{i\cdot\cdot} - \bar y_{\cdots}) - (\bar y_{\cdot j\cdot} - \bar y_{\cdots}) \\
&= \bar y_{ij\cdot} - \bar y_{i\cdot\cdot} - \bar y_{\cdot j\cdot} + \bar y_{\cdots}.
\end{aligned}$$

(The zero-sum constraints on $\hat\tau, \hat\beta, \widehat{\tau\beta}$ kill the other sums.) Hence

$$\hat\mu_{ij} = \hat\mu + \hat\tau_i + \hat\beta_j + \widehat{(\tau\beta)}_{ij} = \bar y_{ij\cdot}, \qquad y_{ijk} = \hat\mu_{ij} + e_{ijk}, \quad e_{ijk} = y_{ijk} - \bar y_{ij\cdot}.$$

2.4 Partition of the total sum of squares

Derivation

$$\begin{aligned} y_{ijk} - \bar{y}_{\cdots} &= (\bar{y}_{ij\cdot} - \bar{y}_{\cdots}) + (y_{ijk} - \bar{y}_{ij\cdot}) \\ &= \underbrace{(\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})}_{\hat{\tau}_i} + \underbrace{(\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots})}_{\hat{\beta}_j} + \underbrace{(\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})}_{\widehat{(\tau\beta)}_{ij}} + \underbrace{(y_{ijk} - \bar{y}_{ij\cdot})}_{e_{ijk}}. \end{aligned}$$

Theorem

$$\begin{aligned} \sum_i \sum_j \sum_k (y_{ijk} - \bar{y}_{\cdots})^2 &= bn \sum_i (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2 + an \sum_j (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots})^2 \\ &\quad + n \sum_i \sum_j (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})^2 + \sum_i \sum_j \sum_k (y_{ijk} - \bar{y}_{ij\cdot})^2, \\ SS_T &= SS_A + SS_B + SS_{AB} + SS_E, \\ abn - 1 &= (a-1) + (b-1) + (a-1)(b-1) + ab(n-1). \end{aligned}$$

Proof. All six cross products vanish:

- any cross product with e_{ijk}: the other factor depends only on (i,j) and $\sum_k (y_{ijk} - \bar{y}_{ij\cdot}) = 0$;
- $\hat{\tau}_i \times \hat{\beta}_j$: $n \sum_i \hat{\tau}_i \sum_j \hat{\beta}_j = 0$;
- $\hat{\tau}_i \times \widehat{(\tau\beta)}_{ij}$: $n \sum_i \hat{\tau}_i \sum_j \widehat{(\tau\beta)}_{ij}$ and $\sum_j (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots}) = b\bar{y}_{i\cdot\cdot} - b\bar{y}_{i\cdot\cdot} - b\bar{y}_{\cdots} + b\bar{y}_{\cdots} = 0$;
- $\hat{\beta}_j \times \widehat{(\tau\beta)}_{ij}$: similarly $\sum_i \widehat{(\tau\beta)}_{ij} = 0$.

df of SS_{AB}: $SS_{AB} = n \sum \sum \widehat{(\tau\beta)}_{ij}^2$ involves ab terms subject to $\sum_i \widehat{(\tau\beta)}_{ij} = 0$ $(j = 1, \ldots, b)$ and $\sum_j \widehat{(\tau\beta)}_{ij} = 0$ $(i = 1, \ldots, a)$. Only $b + (a-1)$ of these $a + b$ conditions are independent (the sum of all row conditions equals the sum of all column conditions), so df $= ab - (a + b - 1) = (a-1)(b-1)$. ■

2.5 Expected mean squares

Theorem

$$\begin{aligned} \mathrm{E}(MS_A) &= \mathrm{E}\Big(\frac{SS_A}{a-1}\Big) = \sigma^2 + \frac{bn \sum_i \tau_i^2}{a-1}, & \mathrm{E}(MS_B) &= \sigma^2 + \frac{an \sum_j \beta_j^2}{b-1}, \\ \mathrm{E}(MS_{AB}) &= \sigma^2 + \frac{n \sum_i \sum_j (\tau\beta)_{ij}^2}{(a-1)(b-1)}, & \mathrm{E}(MS_E) &= \mathrm{E}\Big(\frac{SS_E}{ab(n-1)}\Big) = \sigma^2. \end{aligned}$$

Proof. MS_A**.** Averaging the model, $\bar{y}_{i\cdot\cdot} = \mu + \tau_i + \bar{\varepsilon}_{i\cdot\cdot}$ (since $\sum_j \beta_j = \sum_j (\tau\beta)_{ij} = 0$) and $\bar{y}_{\cdots} = \mu + \bar{\varepsilon}_{\cdots}$. Thus

$$\begin{aligned} \mathrm{E}(SS_A) &= \mathrm{E}\Big[bn \sum_i (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2\Big] = \mathrm{E}\Big[bn \sum_i \big(\tau_i + (\bar{\varepsilon}_{i\cdot\cdot} - \bar{\varepsilon}_{\cdots})\big)^2\Big] \\ &= \mathrm{E}\Big[bn \sum_i \tau_i^2 + bn \sum_i (\bar{\varepsilon}_{i\cdot\cdot} - \bar{\varepsilon}_{\cdots})^2 + 2bn \sum_i \tau_i (\bar{\varepsilon}_{i\cdot\cdot} - \bar{\varepsilon}_{\cdots})\Big] \end{aligned}$$

$$= bn\sum_i \tau_i^2 + bn\,\mathrm{E}\Big[\sum_{i=1}^{a}(\bar\varepsilon_{i\cdot\cdot} - \bar\varepsilon_{\cdot\cdot\cdot})^2\Big] = bn\sum_i \tau_i^2 + bn(a-1)\frac{\sigma^2}{bn},$$

since $\bar\varepsilon_{i\cdot\cdot}$ are iid with variance $\sigma^2/(bn)$. Dividing by $a-1$: $\mathrm{E}(MS_A) = \sigma^2 + \frac{bn}{a-1}\sum_i \tau_i^2$. MS_B is symmetric.

MS_{AB}. $\bar y_{ij\cdot} - \bar y_{i\cdot\cdot} - \bar y_{\cdot j\cdot} + \bar y_{\cdot\cdot\cdot} = (\tau\beta)_{ij} + (\bar\varepsilon_{ij\cdot} - \bar\varepsilon_{i\cdot\cdot} - \bar\varepsilon_{\cdot j\cdot} + \bar\varepsilon_{\cdot\cdot\cdot})$. The $\bar\varepsilon_{ij\cdot}$ form an $a \times b$ table of iid $N(0, \sigma^2/n)$ variables, and the second bracket is exactly the RCBD residual of that table. By the RCBD result of Ch. 4, $\mathrm{E}\sum\sum(\cdot)^2 = (a-1)(b-1)\sigma^2/n$; the cross product has mean 0. Hence

$$\mathrm{E}(SS_{AB}) = n\sum\sum(\tau\beta)_{ij}^2 + n(a-1)(b-1)\frac{\sigma^2}{n} \;\Rightarrow\; \mathrm{E}(MS_{AB}) = \sigma^2 + \frac{n\sum\sum(\tau\beta)_{ij}^2}{(a-1)(b-1)}.$$

MS_E.

$$\mathrm{E}(MS_E) = \frac{1}{ab(n-1)}\mathrm{E}\Big[\sum\sum\sum(y_{ijk} - \bar y_{ij\cdot})^2\Big] = \frac{1}{ab}\sum_i\sum_j \mathrm{E}\Big[\frac{\sum_k(y_{ijk} - \bar y_{ij\cdot})^2}{n-1}\Big] = \frac{1}{ab}\cdot ab\,\sigma^2 = \sigma^2.$$

■

Under each H_0, the corresponding MS and MS_E both estimate σ^2; under H_1 the ratio exceeds 1.

2.6 Extra sum of squares for interaction

Derivation

Full: $y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk} \Rightarrow SS_{E(F)} = \sum\sum\sum(y_{ijk} - \bar y_{ij\cdot})^2$.
Reduced: $y_{ijk} = \mu + \tau_i + \beta_j + \varepsilon_{ijk} \Rightarrow SS_{E(R)} = \sum\sum\sum(y_{ijk} - \bar y_{i\cdot\cdot} - \bar y_{\cdot j\cdot} + \bar y_{\cdot\cdot\cdot})^2$.

$$SS_H = SS_{E(R)} - SS_{E(F)} = n\sum_i\sum_j(\bar y_{ij\cdot} - \bar y_{i\cdot\cdot} - \bar y_{\cdot j\cdot} + \bar y_{\cdot\cdot\cdot})^2 = SS_{AB},$$

$$F_0 = \frac{SS_H/(a-1)(b-1)}{SS_{E(F)}/ab(n-1)} \overset{H_0}{\sim} F_{(a-1)(b-1),\,ab(n-1)}.$$

Proof. $y_{ijk} - \bar y_{i\cdot\cdot} - \bar y_{\cdot j\cdot} + \bar y_{\cdot\cdot\cdot} = (y_{ijk} - \bar y_{ij\cdot}) + (\bar y_{ij\cdot} - \bar y_{i\cdot\cdot} - \bar y_{\cdot j\cdot} + \bar y_{\cdot\cdot\cdot})$; the cross product vanishes because $\sum_k(y_{ijk} - \bar y_{ij\cdot}) = 0$, so $SS_{E(R)} = SS_{E(F)} + SS_{AB}$. (For balanced data the additive-model LSEs are the same marginal means.) ■

2.7 ANOVA table

Source	SS	DF	MS	F_0
A	SS_A	$a-1$	MS_A	MS_A/MS_E
B	SS_B	$b-1$	MS_B	MS_B/MS_E
AB	SS_{AB}	$(a-1)(b-1)$	MS_{AB}	MS_{AB}/MS_E
Error	SS_E	$ab(n-1)$	MS_E	
Total	SS_T	$abn-1$		

Hand formulas:

$$SS_T = \sum\sum\sum y_{ijk}^2 - \frac{y_{\cdot\cdot\cdot}^2}{abn}, \quad SS_A = \frac{\sum_i y_{i\cdot\cdot}^2}{bn} - \frac{y_{\cdot\cdot\cdot}^2}{abn}, \quad SS_B = \frac{\sum_j y_{\cdot j\cdot}^2}{an} - \frac{y_{\cdot\cdot\cdot}^2}{abn},$$

$$SS_{\text{Subtotals}} = \frac{\sum_i \sum_j y_{ij\cdot}^2}{n} - \frac{y_{\cdots}^2}{abn}, \quad SS_{AB} = SS_{\text{Subtotals}} - SS_A - SS_B, \quad SS_E = SS_T - SS_{\text{Subtotals}}.$$

Example: Battery life ANOVA, line by line

$y_{\cdots}^2/abn = 3799^2/36 = 400{,}900.03$, $\sum y^2 = 478{,}547$.

$$\begin{aligned}
SS_T &= 478547 - 400900.03 = 77646.97,\\
SS_{\text{Material}} &= \frac{998^2 + 1300^2 + 1501^2}{12} - 400900.03 = 411583.75 - 400900.03 = 10683.72,\\
SS_{\text{Temp}} &= \frac{1738^2 + 1291^2 + 770^2}{12} - 400900.03 = 440018.75 - 400900.03 = 39118.72,\\
SS_{\text{Subtotals}} &= \frac{539^2 + 229^2 + \cdots + 342^2}{4} - 400900.03\\
&= 460316.25 - 400900.03 = 59416.22,\\
SS_{AB} &= 59416.22 - 10683.72 - 39118.72 = 9613.78,\\
SS_E &= 77646.97 - 59416.22 = 18230.75.
\end{aligned}$$

Source	SS	DF	MS	F_0
Material type	10683.72	2	5341.86	7.91
Temperature	39118.72	2	19559.36	28.97
Interaction	9613.78	4	2403.44	3.56
Error	18230.75	27	675.21	
Total	77646.97	35		

$F_{0.05,4,27} = 2.73$, $F_{0.05,2,27} = 3.35$: interaction ($P = 0.019$) and both main effects are significant. Interaction plot: life decreases with temperature for all materials, but material 3 degrades least at high temperature — the significant interaction means the material effect depends on temperature.

2.8 Multiple comparisons when interaction is present

If interaction is significant, compare one factor's levels *at a fixed level of the other*. For a fixed j, test $H_0 : \mu_{ij} = \mu_{i'j}$ with Fisher's LSD

$$|\bar{y}_{ij\cdot} - \bar{y}_{i'j\cdot}| > LSD = t_{\alpha/2,ab(n-1)}\sqrt{\frac{2MS_E}{n}};$$

for a fixed i, compare $\bar{y}_{ij\cdot}$ and $\bar{y}_{ij'\cdot}$ similarly. (If interaction is not significant, compare main-effect means $\bar{y}_{i\cdot\cdot}$ with $\sqrt{2MS_E/(bn)}$.)

Example: Materials at 70°F

$LSD = t_{0.025,27}\sqrt{2(675.21)/4} = 2.052(18.37) = 37.70$. Cell means at 70°F: $\bar{y}_{12\cdot} = 57.25$, $\bar{y}_{22\cdot} = 119.75$, $\bar{y}_{32\cdot} = 145.75$.

$$|\bar{y}_{12\cdot} - \bar{y}_{22\cdot}| = 62.50 > 37.70, \quad |\bar{y}_{12\cdot} - \bar{y}_{32\cdot}| = 88.50 > 37.70, \quad |\bar{y}_{22\cdot} - \bar{y}_{32\cdot}| = 26.00 < 37.70.$$

At 70°F, types 2 and 3 have the same mean life, and type 1 is significantly lower.

A second example (problem 5.7, pressure × temperature, $n = 2$): F(pressure)= 8.47, F(temp)= 21.59, F(interaction)= 0.97 ($P = 0.47$): no interaction, so main effects are interpreted directly.

2.9 Residual analysis

Residuals $e_{ijk} = y_{ijk} - \bar{y}_{ij\cdot}$; normal Q–Q plot; residuals vs material, vs temperature, vs fitted values $\hat{y}_{ijk} = \bar{y}_{ij\cdot}$.

3 Independence of $\hat{\mu}$ and $\hat{\beta}_j$ (Midterm 2, 2025)

Theorem

In the two-factor model (M1), $\hat{\mu} = \bar{y}_{\cdots}$ and $\hat{\beta}_j = \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}$ are statistically independent.

Proof. Both are linear functions of the normal vector $\mathbf{y}$, so they are jointly normal and independence $\iff$ zero covariance. Since $\bar{y}_{\cdots} = \frac{1}{b}\sum_{j'} \bar{y}_{\cdot j'\cdot}$ and the $\bar{y}_{\cdot j'\cdot}$ are independent (disjoint observations) with variance $\sigma^2/(an)$,

$$\mathrm{Cov}(\bar{y}_{\cdots}, \bar{y}_{\cdot j\cdot}) = \frac{1}{b}\sum_{j'} \mathrm{Cov}(\bar{y}_{\cdot j'\cdot}, \bar{y}_{\cdot j\cdot}) = \frac{1}{b}\mathrm{Var}(\bar{y}_{\cdot j\cdot}) = \frac{\sigma^2}{abn} = \mathrm{Var}(\bar{y}_{\cdots}),$$

$$\mathrm{Cov}(\hat{\mu}, \hat{\beta}_j) = \mathrm{Cov}(\bar{y}_{\cdots}, \bar{y}_{\cdot j\cdot}) - \mathrm{Var}(\bar{y}_{\cdots}) = \frac{\sigma^2}{abn} - \frac{\sigma^2}{abn} = 0.$$

■

4 Comparing Three Models (Midterm 2, 2025)

Model	Error SS	Error df
M1: $y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$	$SS_E^1 = \sum\sum\sum (y_{ijk} - \bar{y}_{ij\cdot})^2$	$ab(n-1)$
M2: $y_{ijk} = \mu + \tau_i + \beta_j + \varepsilon_{ijk}$ $(n > 1)$	$SS_E^2 = \sum\sum\sum (y_{ijk} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})^2$	$abn - a - b + 1$
M3: $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$ $(n = 1)$	$SS_E^3 = \sum\sum (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2$	$(a-1)(b-1)$

For M2: $\hat{\mu} = \bar{y}_{\cdots}$, $\hat{\tau}_i = \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots}$, $\hat{\beta}_j = \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}$, $\hat{y}_{ijk} = \bar{y}_{i\cdot\cdot} + \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}$, $SS_A = bn\sum \hat{\tau}_i^2$ ($a-1$ df), $SS_B = an\sum \hat{\beta}_j^2$ ($b-1$ df).

Test $H_0 : \tau_i = 0$ when the within-cell variation is very small, i.e. $y_{ijk} \approx \bar{y}_{ij\cdot} \approx y_{ij1}$ (the single observation in M3):

$$F_0^{(1)} = \frac{bn\sum_i (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2/(a-1)}{SS_E^1/ab(n-1)}, \qquad SS_E^1 \approx 0 \;\Rightarrow\; F_0^{(1)} \text{ huge},$$

$$F_0^{(2)} = \frac{bn\sum_i (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2/(a-1)}{SS_E^2/(abn - a - b + 1)}, \qquad SS_E^2 \approx n\sum_i\sum_j (y_{ij1} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2 = n\,SS_E^3,$$

$$F_0^{(3)} = \frac{b\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2/(a-1)}{SS_E^3/(a-1)(b-1)}.$$

Since the numerators satisfy $bn\sum(\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2 \approx n \cdot b\sum(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$,

$$F_0^{(2)} \approx \frac{n \cdot b\sum(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2/(a-1)}{n\,SS_E^3/(abn - a - b + 1)} = F_0^{(3)} \cdot \frac{abn - a - b + 1}{(a-1)(b-1)} \;\geq\; F_0^{(3)}.$$

Conclusion: $F_0^{(1)} \gg F_0^{(2)} \geq F_0^{(3)}$. M1 separates pure error from interaction; M2 lumps interaction into error but gains df from replication; M3 has neither replication nor a separate interaction term, so it is the least powerful (and is biased if interaction exists).

5 The General Factorial: Three Factors

Derivation

$$y_{ijkl} = \mu + \tau_i + \beta_j + \gamma_k + (\tau\beta)_{ij} + (\tau\gamma)_{ik} + (\beta\gamma)_{jk} + (\tau\beta\gamma)_{ijk} + \varepsilon_{ijkl},$$
$$SS_T = SS_A + SS_B + SS_C + SS_{AB} + SS_{AC} + SS_{BC} + SS_{ABC} + SS_E,$$

with estimates built hierarchically: $\widehat{(\tau\beta)}_{ij} = \hat{\mu}_{ij} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}_j$, …, $\widehat{(\tau\beta\gamma)}_{ijk} = \hat{\mu}_{ijk} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}_j - \hat{\gamma}_k - \widehat{(\tau\beta)}_{ij} - \widehat{(\tau\gamma)}_{ik} - \widehat{(\beta\gamma)}_{jk}$.

DF: $a-1$, $b-1$, $c-1$, $(a-1)(b-1)$, $(a-1)(c-1)$, $(b-1)(c-1)$, $(a-1)(b-1)(c-1)$, $abc(n-1)$; total $abcn-1$. For fixed effects every F_0 uses MS_E; e.g. $\mathrm{E}(MS_A) = \sigma^2 + \frac{bcn\sum\tau_i^2}{a-1}$, $\mathrm{E}(MS_{ABC}) = \sigma^2 + \frac{n\sum\sum\sum(\tau\beta\gamma)_{ijk}^2}{(a-1)(b-1)(c-1)}$. Note: $n \geq 2$ is required to estimate pure error.

6 Tukey's Test for Additivity (One Observation per Cell)

6.1 The problem

With $n = 1$ and model $y_{ij} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ij}$, the interaction and error cannot be separated: there are ab observations and ab free parameters, so $SS_E = 0$ with 0 df. Assuming no interaction, $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$, and $H_0 : \tau_i = 0$ is tested with $F_0 = MS_A/MS_E$ where MS_E is the residual ($= MS_{AB}$) mean square. Tukey's test checks that assumption using **one** degree of freedom.

6.2 Model

$$y_{ij} = \mu + \tau_i + \beta_j + \gamma\tau_i\beta_j + \varepsilon_{ij}, \qquad \mathrm{E}(y_{ij}) = \mu + \tau_i + \beta_j + \gamma\tau_i\beta_j, \quad \mathrm{Var}(y_{ij}) = \sigma^2,$$

$H_0 : \gamma = 0$ vs $H_1 : \gamma \neq 0$. The interaction is assumed to have the special multiplicative form $(\tau\beta)_{ij} = \gamma\tau_i\beta_j$.

6.3 Estimating μ, τ_i, β_j

Derivation

Since $\sum\tau_i = \sum\beta_j = 0$, the product term averages out:

$$\bar{y}_{\cdot\cdot} = \frac{1}{ab}\sum_i\sum_j(\mu + \tau_i + \beta_j + \gamma\tau_i\beta_j + \varepsilon_{ij}) = \mu + \bar{\varepsilon}_{\cdot\cdot}$$
$$\Rightarrow\ \mathrm{E}(\bar{y}_{\cdot\cdot}) = \mu, \quad \hat{\mu} = \bar{y}_{\cdot\cdot} \text{ unbiased},$$
$$\bar{y}_{i\cdot} = \frac{1}{b}\sum_j(\mu + \tau_i + \beta_j + \gamma\tau_i\beta_j + \varepsilon_{ij}) = \mu + \tau_i + \bar{\varepsilon}_{i\cdot}$$
$$\Rightarrow\ \hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} \text{ unbiased}; \quad \text{similarly } \hat{\beta}_j = \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot} \text{ unbiased}.$$

6.4 Least squares estimator of γ

Derivation

$S(\theta) = \sum_i \sum_j (y_{ij} - \mu - \tau_i - \beta_j - \gamma\tau_i\beta_j)^2 = \sum\sum \varepsilon_{ij}^2$. Write $S_{ij} := y_{ij} - \mu - \tau_i - \beta_j - \gamma\tau_i\beta_j$.

$$\frac{\partial S}{\partial \gamma}\Big|_{\hat\theta} = 0 \;\Rightarrow\; \sum_i \sum_j \tau_i\beta_j S_{ij} = 0 \;\Rightarrow$$

$$\sum\sum \tau_i\beta_j y_{ij} - \mu \underbrace{\sum\sum \tau_i\beta_j}_{0} - \underbrace{\sum\sum \tau_i^2\beta_j}_{0} - \underbrace{\sum\sum \tau_i\beta_j^2}_{0} - \gamma \sum \tau_i^2 \sum \beta_j^2 = 0.$$

Hence (plugging in the estimated effects)

$$\hat\gamma = \frac{\sum_i \sum_j \hat\tau_i \hat\beta_j y_{ij}}{\sum_i \hat\tau_i^2 \sum_j \hat\beta_j^2} = \frac{ab \sum\sum \hat\tau_i \hat\beta_j y_{ij}}{SS_A \cdot SS_B},$$

because $SS_A = b\sum_i \hat\tau_i^2$ and $SS_B = a\sum_j \hat\beta_j^2$.

6.5 Sum of squares for non-additivity

Theorem

$$SS_N = \sum_i \sum_j \hat\gamma^2 \hat\tau_i^2 \hat\beta_j^2 = \frac{\left(\sum\sum \hat\tau_i \hat\beta_j y_{ij}\right)^2}{\sum \hat\tau_i^2 \sum \hat\beta_j^2} = \frac{\left[\sum_{i=1}^a \sum_{j=1}^b (\bar y_{i\cdot} - \bar y_{\cdot\cdot})(\bar y_{\cdot j} - \bar y_{\cdot\cdot}) y_{ij}\right]^2}{\sum_{i=1}^a (\bar y_{i\cdot} - \bar y_{\cdot\cdot})^2 \sum_{j=1}^b (\bar y_{\cdot j} - \bar y_{\cdot\cdot})^2}$$

$$= \frac{ab\left(\sum\sum \hat\tau_i \hat\beta_j y_{ij}\right)^2}{SS_A\, SS_B}.$$

SS_N has 1 df; $SS_E = SS_{\text{Residual}} - SS_N$ with $(a-1)(b-1) - 1$ df, where $SS_{\text{Residual}} = \sum\sum (y_{ij} - \bar y_{i\cdot} - \bar y_{\cdot j} + \bar y_{\cdot\cdot})^2$. $SS_N = SS_H = SS_{E(R)} - SS_{E(F)}$ is the extra SS for γ.

Proof. $SS_N = \hat\gamma^2 \sum_i \hat\tau_i^2 \sum_j \hat\beta_j^2 = \frac{(\sum\sum \hat\tau_i\hat\beta_j y_{ij})^2}{(\sum \hat\tau_i^2 \sum \hat\beta_j^2)^2} \sum \hat\tau_i^2 \sum \hat\beta_j^2$, which simplifies to the stated forms. ■

6.6 Distribution of SS_N under H_0

Unconditional (oracle) version. If τ_i, β_j were known, $\tilde\gamma = \frac{\sum\sum \tau_i\beta_j y_{ij}}{\sum \tau_i^2 \sum \beta_j^2}$ satisfies

$$\mathrm{E}(\tilde\gamma) = \frac{\sum\sum \tau_i\beta_j(\mu + \tau_i + \beta_j + \gamma\tau_i\beta_j)}{\sum \tau_i^2 \sum \beta_j^2} = \gamma,$$

$$\mathrm{Var}(\tilde\gamma) = \Big(\frac{1}{\sum \tau_i^2 \sum \beta_j^2}\Big)^2 \sum\sum \tau_i^2\beta_j^2\sigma^2 = \frac{\sigma^2}{\sum \tau_i^2 \sum \beta_j^2},$$

so under H_0, $\tilde\gamma \sim N(0, \mathrm{Var}\tilde\gamma)$ and $\tilde\gamma^2/\mathrm{Var}(\tilde\gamma) \sim \chi_1^2$.

With estimated effects. Condition on $\hat\tau, \hat\beta$. Under H_0,

$$\mathrm{E}(\hat\gamma \mid \hat\tau, \hat\beta, H_0) = \frac{ab \sum\sum \hat\tau_i\hat\beta_j(\mu + \tau_i + \beta_j)}{SS_A\, SS_B} = 0$$

(the weights $c_{ij} = \hat\tau_i\hat\beta_j$ have zero row and column sums, so they annihilate any additive table). Moreover $\sum\sum c_{ij}y_{ij} = \sum\sum c_{ij}e_{ij}$ depends only on the residuals, which are independent of the row and column means under normality; hence

$$\mathrm{Var}(\hat\gamma \mid \hat\tau, \hat\beta) = \Big(\frac{1}{\sum\hat\tau_i^2 \sum\hat\beta_j^2}\Big)^2 \sum\sum \hat\tau_i^2\hat\beta_j^2\sigma^2 = \frac{\sigma^2}{\sum\hat\tau_i^2\sum\hat\beta_j^2} = \frac{ab\sigma^2}{SS_A\,SS_B} =: \widehat{\mathrm{Var}}(\hat\gamma).$$

Therefore

$$\frac{\hat\gamma^2}{\widehat{\mathrm{Var}}(\hat\gamma)} = \frac{(ab)^2(\sum\sum\hat\tau_i\hat\beta_j y_{ij})^2}{SS_A^2\,SS_B^2}\cdot\frac{SS_A\,SS_B}{ab\,\sigma^2} = \frac{ab(\sum\sum\hat\tau_i\hat\beta_j y_{ij})^2}{SS_A\,SS_B\,\sigma^2} = \frac{SS_N}{\sigma^2} \overset{H_0}{\sim} \chi^2_1,$$

conditionally on $(\hat\tau, \hat\beta)$, and since this conditional law does not depend on $(\hat\tau, \hat\beta)$, also unconditionally.

Theorem: Cochran-type lemma

If Q, Q_1, Q_2 are quadratic forms with $Q = Q_1 + Q_2$, $Q \sim \chi^2_n$, $Q_1 \sim \chi^2_{n_1}$ and $Q_2 \geq 0$, then $Q_2 \sim \chi^2_{n-n_1}$ and $Q_1 \perp\!\!\!\perp Q_2$.

Apply it with $Q = SS_{\text{Residual}}/\sigma^2 \sim \chi^2_{(a-1)(b-1)}$ (under H_0, the additive model is true), $Q_1 = SS_N/\sigma^2 \sim \chi^2_1$, $Q_2 = SS_E/\sigma^2$:

$$\frac{SS_E}{\sigma^2} \sim \chi^2_{(a-1)(b-1)-1}, \quad SS_N \perp\!\!\!\perp SS_E$$

$$\Rightarrow \quad F_0 = \frac{MS_N}{MS_E} = \frac{SS_N/1}{SS_E/[(a-1)(b-1)-1]} \overset{H_0}{\sim} F_{1,(a-1)(b-1)-1}.$$

Reject $H_0 : \gamma = 0$ if $F_0 > F_{\alpha,1,(a-1)(b-1)-1}$.

Source	SS	DF	MS	F_0
A	SS_A	$a-1$	MS_A	MS_A/MS_E
B	SS_B	$b-1$	MS_B	MS_B/MS_E
Nonadditivity	SS_N	1	MS_N	MS_N/MS_E
Error	SS_E	$(a-1)(b-1)-1$	MS_E	
Total	SS_T	$ab-1$		

Example: Impurity data (Table 5.10), temperature ($a = 3$) × pressure ($b = 5$), $n = 1$

Temp	25	30	35	40	45	$y_{i\cdot}$
100	5	4	6	3	5	23
125	3	1	4	2	3	13
150	1	1	3	1	2	8
$y_{\cdot j}$	9	6	13	6	10	44

$\bar y_{\cdot\cdot} = 44/15 = 2.933$; $\hat\tau = (4.6, 2.6, 1.6) - 2.933 = (1.667, -0.333, -1.333)$, $\sum\hat\tau_i^2 = 4.667$; $\hat\beta = (3, 2, 4.333, 2, 3.333) - 2.933 = (0.067, -0.933, 1.400, -0.933, 0.400)$, $\sum\hat\beta_j^2 = 3.867$.

$$SS_A = 5(4.667) = 23.33, \quad SS_B = 3(3.867) = 11.60,$$
$$SS_{\text{Residual}} = 166 - \tfrac{44^2}{15} - 23.33 - 11.60 = 2.00,$$
$$\sum\sum\hat\tau_i\hat\beta_j y_{ij} = 1.333,$$

$$SS_N = \frac{1.333^2}{4.667 \times 3.867} = \frac{1.778}{18.04} = 0.0985,$$
$$SS_E = 2.00 - 0.0985 = 1.9015 \text{ (7 df)}, \quad MS_E = 0.2716,$$
$$F_0 = \frac{0.0985}{0.2716} = 0.36 < F_{0.05,1,7} = 5.59.$$

No evidence of interaction; the additive analysis ($F_{\text{temp}} = 11.667/0.25 = 46.67$, $F_{\text{press}} = 2.9/0.25 = 11.6$) is justified.